Module 5, Lesson 2

AI 101 for Mechanical Engineers

Scaling and encoding

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Outline

1. Feature Scaling Transformations
2. Categorical Data Encoding
3. Operational Constraints
4. Synthesis and Summary

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1. Feature Scaling Transformations

Purpose: Mathematical methods for mapping continuous engineering data to defined scales.

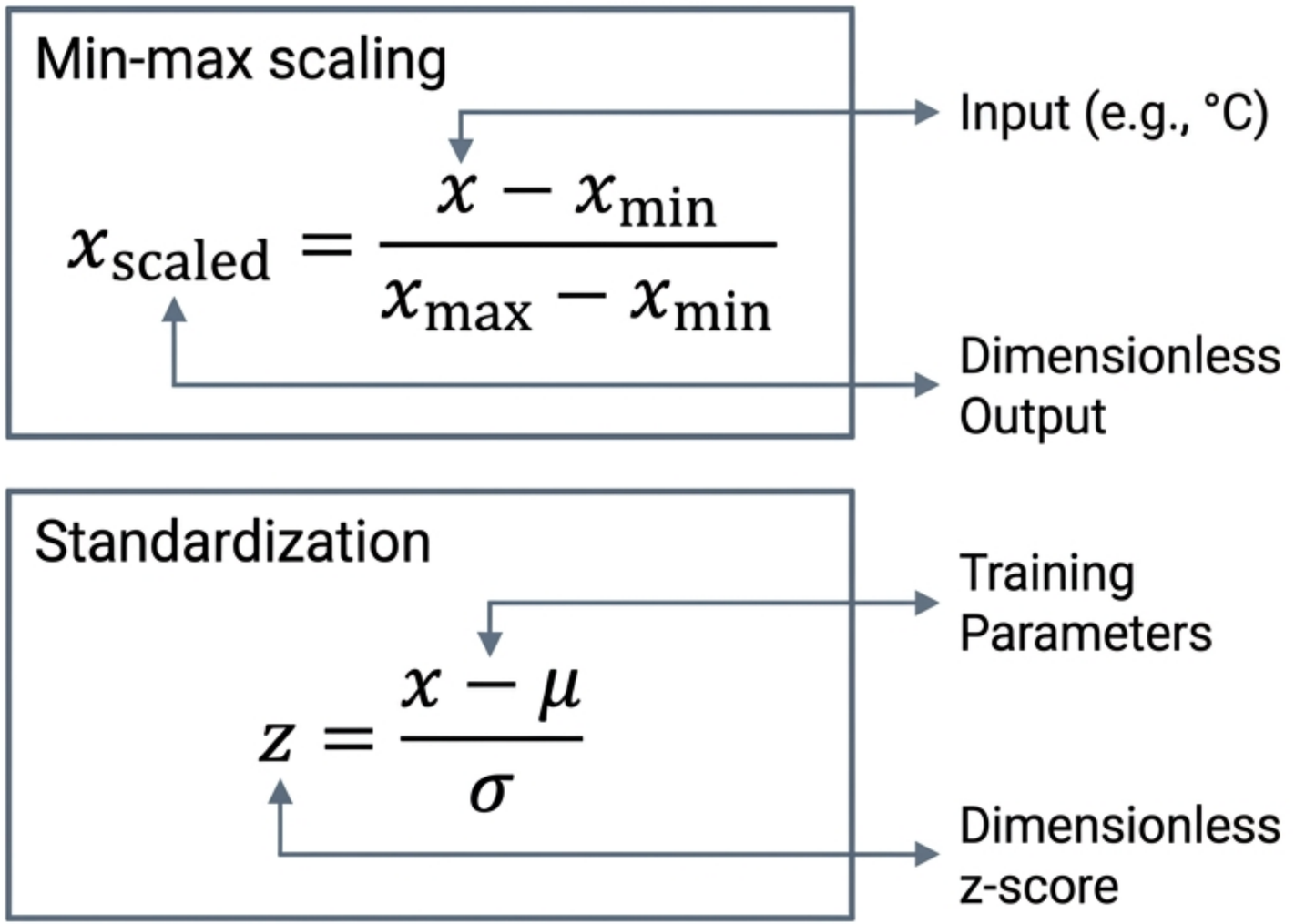
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Continuous Feature Scaling Equations

Transform feature magnitudes to a defined numerical scale.

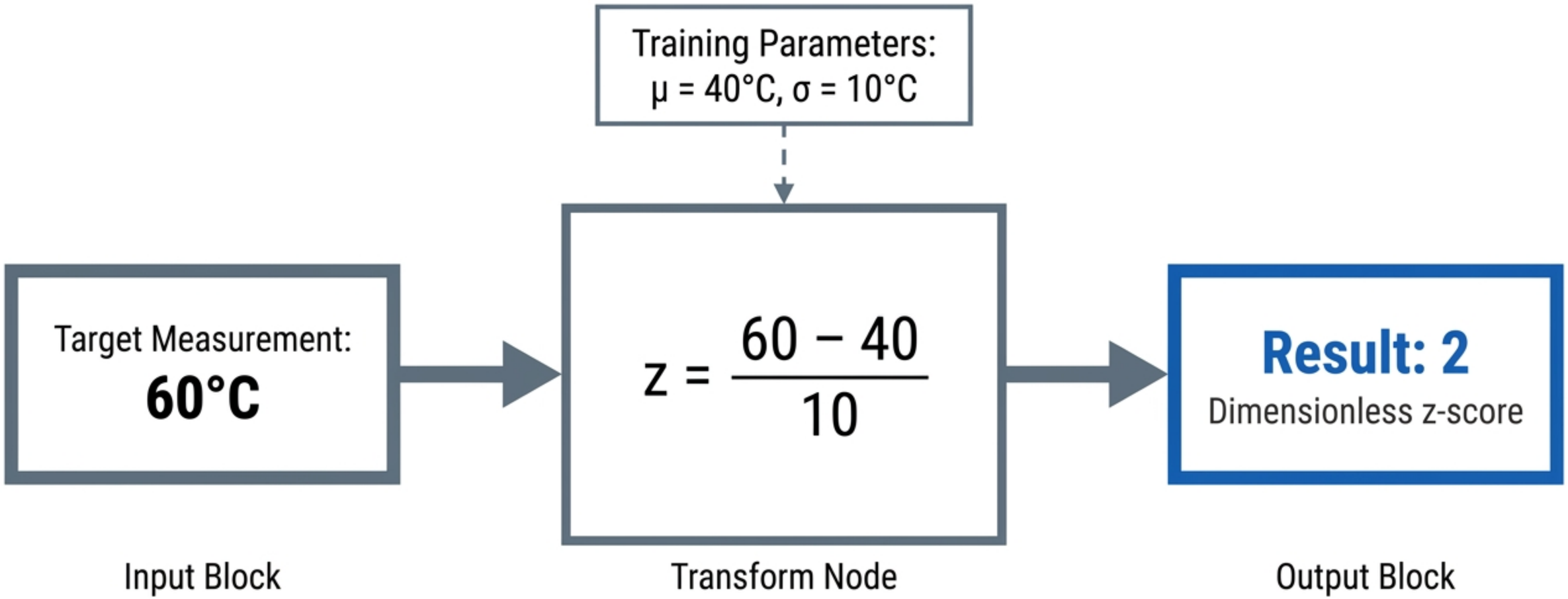
Specify exact mathematical rules instead of overloaded terms like normalization.

Retain original physical units for post-processing interpretation.



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Calculating Standardized Engineering Values



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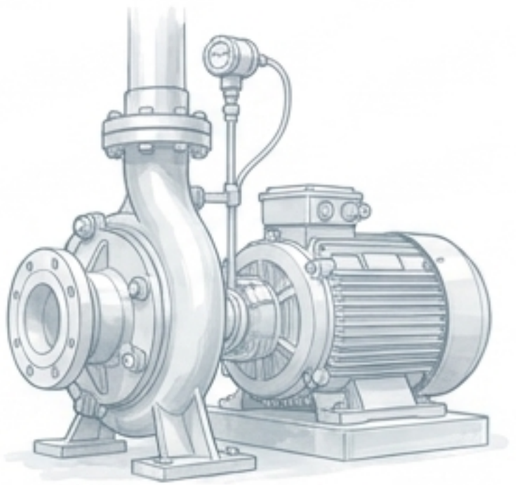
2. Categorical Data Encoding

Purpose: Converting non-numerical categorical states into mathematical representations.

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Representing Categorical Engineering Features

Assigning integers (1, 2, 3) imposes unjustified numerical ordering.
One-hot encoding generates distinct binary indicator columns.



Machine A	**1**	0	0
Machine B	0	**1**	0
Machine C	0	0	**1**

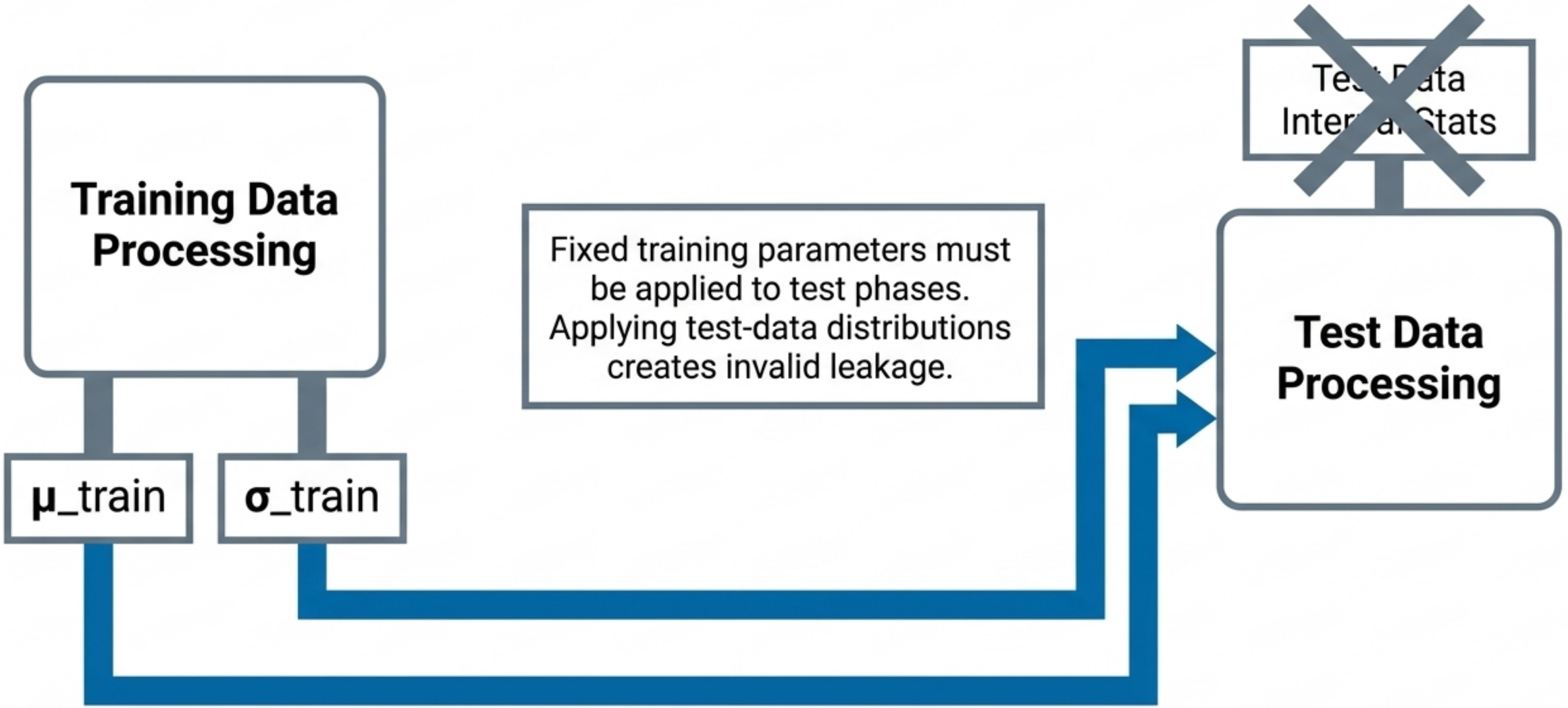
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3. Operational Constraints

Purpose: Preventing statistical leakage and managing edge-case features.

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Preventing Statistical Data Leakage

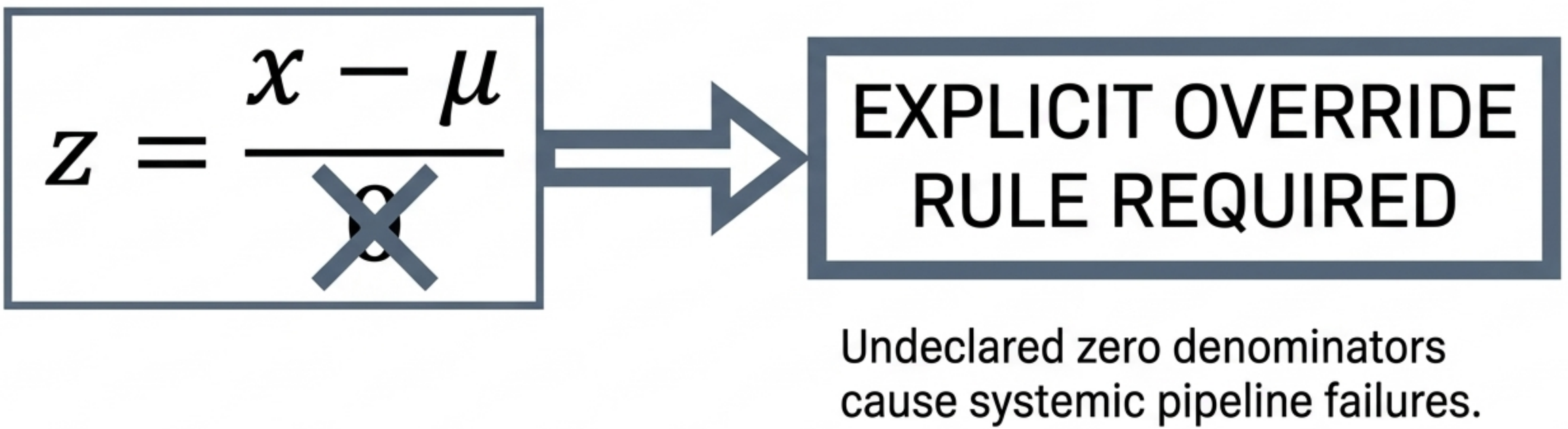


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Managing Zero-Variance Features

A constant engineering feature possesses a standard deviation of zero.

Standardization mathematically requires dividing by the standard deviation.



Undeclared zero denominators cause systemic pipeline failures.

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4. Synthesis and Summary

Purpose: Consolidated review of transformation methodologies and rules.

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Transformation Concept Reference Matrix

Transform	Training Quantities	Example Output
Min-Max	Bounds 0–100°C	25°C becomes 0.25
Standardization	Mean 40°C, SD 10°C	60°C becomes 2
One-Hot Encoding	Known categories	Indicator columns

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Key Points



Section 1: Feature scaling transforms magnitudes while retaining original unit interpretations.



Section 2: One-hot encoding represents categories without imposing false mathematical hierarchies.



Section 3: Fixed training parameters and constant-feature rules ensure pipeline integrity.

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